Supplementary material: Lyman-α Warm Dark Matter Limits Compared with Non-Resonant Sterile-Neutrino Production

HEP-Genesis-Agent session record
Project: ramp-combo-lyman-wdm

2026-10-19

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `54403ba5`
- Started: 2026-10-19 14:31:14; report generated: 2026-10-19 14:42:05
- Project: ramp-combo-lyman-wdm (`/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm`)
- Compute execution site(s) selected in the session: local
- 1 user turn(s), 97 tool call(s) (0 returned errors), 1 figure(s), 12 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
report	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
coder	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`

Server	Transport	Location
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`serve-sbbn.sh list_sterile_nu_skills`	`{}`	`{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...`
2	`serve-hydro.sh list_observables`	`{}`	`{ "status": "success", "message": "14 observables available: GSMF, HMF, fGas, Pk-ratio, CSFR, CGD, CGED, CPP, CTP, CEP, CEEP, CMP, CYP, Pk_GO.", "files": [], "metadata": { "observables": {...`
3	`hep-research-aid list_agent_files`	`{}`	`[]`
4	`hep-research-aid list_input_files`	`{}`	`{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }`
5	`hep-research-aid list_datasets`	`{}`	`{'count': 0, 'datasets': []}`
6	`hep-research-aid list_files`	`{}`	`Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...`
7	`hep-research-aid get_dispatch`	`{}`	`Dispatch: local execution.`

#	Tool	Arguments	Outcome
8	local search_arxiv_ abstracts	{"query": "Lyman-alpha forest warm dark matter thermal relic mass lower limit", "max_results": 8}	[1107.3837v2] The Lyman-alpha forest in a blazar-heated Universe Abstract: It has been realised only recently that TeV emission from blazars can significantly heat the intergalactic medium (IGM) by pa...
9	local search_arxiv_ abstracts	{"query": "non-resonant Dodelson-Widrow sterile neutrino Lyman-alpha mass mapping thermal relic", "max_results": 6}	[0507333v2] UHE neutrino damping in a thermal gas of relic neutrinos Abstract: We present a calculation of the damping of an ultra-energetic (UHE) cosmic neutrino travelling through the thermal gas ...
10	local search_arxiv_ abstracts	{"query": "sterile neutrino dark matter X-ray Lyman-alpha combined constraints exclusion Dodelson-Widrow", "max_ results": 6}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
11	hep-research-aid search_arxiv	{"query": "new constraints on the free-streaming of warm dark matter from intermediate and small scale Lyman-alpha forest", "max_ results": 5}	Found 5 papers matching query: 'new constraints on the free-streaming of warm dark matter from intermediate and small scale Lyman-alpha forest' (source: arXiv API) Paper 1: arXiv ID: 1107.3837v2 Titl...
12	local search_arxiv_ abstracts	{"max_results": 6, "query": "Unveiling dark matter free streaming smallest scales high redshift Lyman-alpha forest"}	[2309.04533v2] Unveiling Dark Matter free-streaming at the smallest scales with high redshift Lyman-alpha forest Abstract: This study introduces novel constraints on the free-streaming of thermal reli...

#	Tool	Arguments	Outcome
13	local search_arxiv_ abstracts	{"max_results": 6, "query": "hints neutrino mass warm dark matter eBOSS XQ-100 Lyman-alpha"}	[2203.16563v1] Could fresh lava be (warm) dark matter? Abstract: Dark matter models can be classified according to their impact on the properties of galaxies, including cold dark matter (CDM), warm da...
14	local get_arxiv_ metadata	{"arxiv_id": "2309.04533"}	arXiv ID: 2309.04533 Title: Unveiling Dark Matter free-streaming at the smallest scales with high redshift Lyman-alpha forest Authors: Vid Ir?i?, Matteo Viel, Martin G. Haehnelt, James S. Bolton, Marg...
15	local get_arxiv_ metadata	{"arxiv_id": "1702.01764"}	arXiv ID: 1702.01764 Title: New Constraints on the free-streaming of warm dark matter from intermediate and small scale Lyman-α forest data Authors: Vid Ir?i?, Matteo Viel, Martin G. Haehnelt, James...
16	local get_arxiv_ metadata	{"arxiv_id": "2209.14220"}	arXiv ID: 2209.14220 Title: New Constraints on Warm Dark Matter from the Lyman-α Forest Power Spectrum Authors: Bruno Villasenor, Brant Robertson, Piero Madau, Evan Schneider Year: 2022 Abstract: Th...
17	local get_arxiv_ metadata	{"arxiv_id": "1306.2314"}	arXiv ID: 1306.2314 Title: Warm Dark Matter as a solution to the small scale crisis: new constraints from high redshift Lyman-alpha forest data Authors: M. Viel, G. D. Becker, J. S. Bolton, M. G. Haeh...

#	Tool	Arguments	Outcome
18	local get_arxiv_ metadata	{"arxiv_id": "1705.01837"}	arXiv ID: 1705.01837 Title: Sterile neutrinos in cosmology Authors: Kevork N. Abazajian Year: 2017 Abstract: Sterile neutrinos are natural extensions to the standard model of particle physics in neutr...
19	local get_arxiv_ metadata	{"arxiv_id": "1911.09073"}	arXiv ID: 1911.09073 Title: Hints, neutrino bounds and WDM constraints from SDSS DR14 Lyman-α and Planck full-survey data Authors: Nathalie Palanque-Delabrouille, Christophe Yèche, Nils Schöneberg, ...
20	serve-sbbn.sh describe_sterile_ nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\ nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
21	serve-sbbn.sh list_sterile_nu_ skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
22	serve-sbbn.sh load_sterile_nu_ skill	{"name": "kev-sterile-dm-xray"}	{ "status": "ok", "files": [], "message": "Loaded skill 'kev-sterile-dm-xray'. Follow these instructions.", "metadata": { "instructions": "--\nname: kev-sterile-dm-xray\ ndescription: keV ...

#	Tool	Arguments	Outcome
23	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 7100, "sin2_2theta": 5e-11}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 7.1e+03 eV, sin\u00b22\u03b8 = 5e-11\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff ...
24	local download_arxiv_full_text	{"arxiv_id": "astro-ph/0501562", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/astro-ph_0501562.txt
25	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [7100, 20000, 29000, 41000, 46000], "sin2_2thetas": [1e-09], "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_nu_grid_DW_4bf99b.csv"], "message": "Predicted 5 points [DW] ...
26	local download_arxiv_full_text	{"arxiv_id": "2309.04533", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/2309.04533.txt
27	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...

#	Tool	Arguments	Outcome
28	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1000, 1500, 2000, 3000, 5000, 7100, 10000, 15000, 20000, 22000, 29000, 35000, 41000, 46000, 50000], "sin2_2thetas": [1e-13, 3e-13, 1e-12, 3e-12, 1e-11, 3e-11, 1e-10, 3e-10, 1e-09, 3e-09, 1e-08, 3e-08, 1e-0...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_nu_grid_DW_8d35ed.csv"], "message": "Predicted 195 points [D...
29	local download_arxiv_full_text	{"arxiv_id": "1702.01764", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/1702.01764.txt
30	hep-research-aid load_table_data	{"filename": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_nu_grid_DW_8d35ed.csv"}	Error: could not convert string 'm_s_eV' to float64 at row 0, column 1.
31	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/astro-ph_0501562.txt", "start": 0, "length": 12000}	[chars 0-12000 of 50343] (more available, next start=12000) Title: Constraining Warm Dark Matter candidates including sterile neutrinos and light gravitinos with WMAP and the Lyman-alpha forest Autho...
32	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/2309.04533.txt", "start": 0, "length": 6000}	[chars 0-6000 of 106880] (more available, next start=6000) Title: Unveiling Dark Matter free-streaming at the smallest scales with high redshift Lyman-alpha forest Authors: Vid Ir?i?, Matteo Viel, Ma...

#	Tool	Arguments	Outcome
33	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/1702.01764.txt", "start": 0, "length": 5000}	[chars 0-5000 of 71193] (more available, next start=5000) Title: New Constraints on the free-streaming of warm dark matter from intermediate and small scale Lyman-α forest data Authors: Vid Ir?i?, ...
34	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_nu_grid_DW_8d35ed.csv"}	m_s_eV,sin2_2theta, delta_neff,neff,omega_s_h2,Yp,DH_x1e5,Li7H_x1e10,chi2_bbn,delta_chi2_bbn,lifetime_s, free_streaming_kpc,bbn_excluded_95,overcloses, bbn_treatment_valid 1000, 1e-13,1.28919e-07,3.044, 1....
35	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [7100, 20000, 29000, 41000, 46000], "sin2_2thetas": [2.545e-10, 3.946e-11, 2.022e-11, 1.084e-11, 8.81e-12], "combine": "zip", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_nu_grid_DW_e67bfc.csv"], "message": "Predicted 5 points [DW] ...
36	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100, "sin2_2theta": 2.545e-10}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 2.5e-10\n\nE_\u03b3 = 3.55 keV\nDec...
37	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 20000, "sin2_2theta": 3.946e-11}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 3.9e-11\n\nE_\u03b3 = 10 keV\nDecay ...

#	Tool	Arguments	Outcome
38	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 29000, "sin2_2theta": 2.022e-11}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 29 keV, sin\u00b22\u03b8 = 2e-11\n\nE_\u03b3 = 14.5 keV\nDecay ...
39	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 41000, "sin2_2theta": 1.084e-11}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 41 keV, sin\u00b22\u03b8 = 1.1e-11\n\nE_\u03b3 = 20.5 keV\nDeca...
40	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 46000, "sin2_2theta": 8.81e-12}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 46 keV, sin\u00b22\u03b8 = 8.8e-12\n\nE_\u03b3 = 23 keV\nDecay ...
41	local cite	{"arxiv_id": "astro-ph/0501562", "note": "[\"There is a simple one-to-one correspondence between the masses of thermal WDM particles and sterile neutrinos for which the e?ect on the matter distribution and thus the trans...	Cited: Constraining Warm Dark Matter candidates including sterile neutrinos and light gravitinos with WMAP and the Lyman-alpha forest (\cite{arxivastroph0501562})
42	local cite	{"arxiv_id": "2309.04533", "note": "[\"to establish lower limits on the mass of a thermal relic WDM particle of5.7 keV (at 95% C.L.)\", \"lowers the bound on the WDM mass to4.1 keV\", \"At the smallest scales, the primar...	Cited: Unveiling Dark Matter free-streaming at the smallest scales with high redshift Lyman-alpha forest (\cite{arxiv230904533})

#	Tool	Arguments	Outcome
43	local cite	{"arxiv_id": "1702.01764", "note": "[\"the combined analysis gives our best and signi?cantly strengthened lower limit of 5.3 keV (all 2 σ C.L.)\", \"reduces the lower limit for the combined analysis to 3.5 keV\"]", "bib_...	Cited: New Constraints on the free-streaming of warm dark matter from intermediate and small scale Lyman-α forest data (\cite{arxiv170201764})
44	local get_arxiv_ metadata	{"arxiv_id": "2209.14220"}	arXiv ID: 2209.14220 Title: New Constraints on Warm Dark Matter from the Lyman-α Forest Power Spectrum Authors: Bruno Villasenor, Brant Robertson, Piero Madau, Evan Schneider Year: 2022 Abstract: Th...
45	local get_arxiv_ metadata	{"arxiv_id": "1911.09073"}	arXiv ID: 1911.09073 Title: Hints, neutrino bounds and WDM constraints from SDSS DR14 Lyman-α and Planck full-survey data Authors: Nathalie Palanque-Delabrouille, Christophe Yèche, Nils Schöneberg, ...
46	local get_arxiv_ metadata	{"arxiv_id": "1306.2314"}	arXiv ID: 1306.2314 Title: Warm Dark Matter as a solution to the small scale crisis: new constraints from high redshift Lyman-alpha forest data Authors: M. Viel, G. D. Becker, J. S. Bolton, M. G. Haeh...
47	serve-sbbn.sh predict_sterile_ nu_grid	{"masses_eV": [22000, 29000, 41000, 46000, 50000], "sin2_ 2thetas": [1e-12, 1e-11], "mechanism": "DW", "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/runs/ 261019_143114"}	{ "status": "ok", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/ runs/261019_143114/ sterile_nu_grid_DW_ 10ad91.csv"], "message": "Predicted 10 points [DW...

#	Tool	Arguments	Outcome
48	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 1000, "m_max_eV": 100000, "theta_min": 1e-14, "theta_max": 1e-06, "mechanism": "DW", "n_points": 80, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/scan_DW_d94a06.csv", "/Users/nramachandra/hep-genesis-testing/project...
49	board post_finding	{"claim": "On the sterile-nu server's DW relic curve (Omega_s h^2 = 0.12), the mixing is sin^2 2theta = 2.545e-10, 3.946e-11, 2.022e-11, 1.084e-11 and 8.81e-12 at m_s = 7.1, 20, 29, 41 and 46 keV. The corresponding radia...	Recorded finding f-001.
50	board post_finding	{"claim": "At the DW relic mixing, the line energies are E = 3.55, 10, 14.5, 20.5 and 23 keV for m_s = 7.1, 20, 29, 41 and 46 keV. The whole-halo M31 fluxes are 1.75e-3, 1.71e-2, 3.88e-2, 8.31e-2 and 0.107 ph/s/cm^2. The...	Recorded finding f-002.
51	local search_arxiv_abstracts	{"query": "NuSTAR INTEGRAL sterile neutrino decay line limits 10-50 keV mixing angle", "max_results": 6}	[1606.02684v1] Neutrino Oscillations With Two Sterile Neutrinos Abstract: This work estimates the probability of $$\mu$$ to e neutrino oscillation with two sterile neutrinos using a 5x5 U-matrix, an ext...
52	local search_arxiv_abstracts	{"query": "Lyman-alpha mixed cold warm dark matter fraction bound sterile neutrino", "max_results": 6}	[0701585v1] Thermal and chemical evolution of the primordial clouds in warm dark matter models with keV sterile neutrinos in one-zone approximation Abstract: We follow the evolution of the baryonic ...

#	Tool	Arguments	Outcome
53	local get_arxiv_ metadata	{"arxiv_id": "1609.00667"}	arXiv ID: 1609.00667 Title: (Almost) Closing the νMSM Sterile Neutrino Dark Matter Window with NuSTAR Authors: Kerstin Perez, Kenny C. Y. Ng, John F. Beacom, Cora Hersh, Shunsaku Horiuchi, Roman Kri...
54	local get_arxiv_ metadata	{"arxiv_id": "0812.0010"}	arXiv ID: 0812.0010 Title: Lyman-alpha constraints on warm and on warm-plus-cold dark matter models Authors: Alexey Boyarsky, Julien Lesgourgues, Oleg Ruchayskiy, Matteo Viel Year: 2008 Abstract: We...
55	local cite	{"arxiv_id": "0812.0010", "bib_path": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/ runs/261019_143114/ references.bib", "note": "[\"we find that any mass value is allowed when F_WDM < 0.6 (freque...	Cited: Lyman-alpha constraints on warm and on warm-plus-cold dark matter models (\ cite{arxiv08120010})
56	board post_finding	{"claim": "On the sterile-nu server (DW mechanism), Omega_s h^2 scales linearly with sin^2 2theta and roughly as m_s^ 1.8. Setting Omega_s h^2 = 0.12 gives a DW relic curve of sin^2 2theta $\approx$ 8.7e-9 (1 keV), 2.5e-9 (2 keV)...	Recorded finding f-003.
57	board post_finding	{"claim": "DW fractions f = Omega_s h^2 / 0.12 from the server: at 7.1 keV and sin^ 2 2theta = 5e-11, f = 0.196 (Omega_s h^2 = 0.0236). At sin^2 2theta = 1e-12, f = 0.025 (20 keV), 0.030 (22 keV), 0.049 (29 keV), 0.092 (4...	Recorded finding f-004.

#	Tool	Arguments	Outcome
58	board post_finding	{"claim": "Verified from the text of Viel+2005 (astro-ph/0501562, Eq. 5): m_s(DW) = 4.43 keV (m_x/1 keV)^{4/3} (0.25*0.7^2/omega_x)^{1/3}, normalized at omega_x = 0.1225; the survey's 4.46 keV prefactor is not in the pap...	Recorded finding f-005.
59	board post_finding	{"claim": "Verified Lyman-alpha thermal-relic WDM limits: Viel+2013 (HIRES/MIKE, z>4) m_x>3.3 keV (2 sigma; 1, 2, 2.5 keV disfavoured at about 9, 4, 3 sigma); Irsic+2017 XQ-100 alone 1.4 keV, HIRES/MIKE 4.1 keV, combined...	Recorded finding f-006.
60	board post_question	{"text": "Do any post-2023 Lyman-alpha WDM limits exist (DESI DR1 P1D, eBOSS updates, or Rogers/Peiris emulator re-analyses), and do 2209.14220, 1911.09073 or 1306.2314 quote their own DW or NRP sterile-equivalent masses...	Recorded question q-05.
61	serve-sbbn.sh plot_sterile_nu_exclusion	{"scan_file": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/scan_DW_d94a06.npz", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/26101...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/exclusion_scan_DW_d94a06.png", "/Users/nramachandra/hep-genesis-testi... [Fig. 10]
62	local search_arxiv_abstracts	{"query": "Lyman-alpha forest warm dark matter thermal relic mass lower limit 2024 DESI", "max_results": 6}	[2404.03004v2] Validation of the DESI 2024 Lyα forest BAO analysis using synthetic datasets Abstract: The first year of data from the Dark Energy Spectroscopic Instrument (DESI) contains the largest...

#	Tool	Arguments	Outcome
63	board post_finding	{"claim": "The mixed cold-plus-warm Lyman-alpha bound of Boyarsky+2009 (arXiv:0812.0010), with masses expressed as non-resonantly produced sterile-neutrino masses, allows any mass in 5 keV < m_NRP < infinity when F_WDM <...	Recorded finding f-007.
64	board post_finding	{"claim": "NuSTAR Galactic Center observations (Perez+2017, arXiv: 1609.00667) give, by the abstract's own description, the strongest X-ray line limits on sterile-neutrino mixing over most of 10-50 keV in mass, improving ...	Recorded finding f-008.
65	board post_question	{"text": "What are the numerical NuSTAR (1609.00667, Roach+2020 and later) upper limits on sin^ 2 2theta at m_s = 29, 41 and 46 keV? Are they below the DW relic mixings of 2.0e-11, 1.1e-11 and 8.8e-12?"}	Recorded question q-06.
66	board post_dead_end	{"what": "A search_arxiv_ abstracts query for a post-2023 DESI Lyman-alpha thermal-relic WDM limit", "why": "The query returned only DESI 2024 BAO and sample papers. No DESI DR1 P1D-based WDM mass limit came up, so no new...	Recorded dead end x-01.
67	board post_finding	{"claim": "I made a DW exclusion plot over m_s = 1-100 keV and sin^2 2theta = 1e-14 to 1e-6, from an 80x80 scan (6400 points): 3092 points overclose and 968 are BBN-excluded. The plot shows the overclosure region, the Om...	Recorded finding f-009.

#	Tool	Arguments	Outcome
68	local search_arxiv_ abstracts	{"query": "one-dimensional flux power spectrum warm dark matter 2024", "max_ results": 10}	[2203.16563v1] Could fresh lava be (warm) dark matter? Abstract: Dark matter models can be classified according to their impact on the properties of galaxies, including cold dark matter (CDM), warm da...
69	local download_arxiv_ full_text	{"arxiv_id": "1609.00667", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/runs/ 261019_143114"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/ runs/261019_143114/ 1609.00667.txt
70	serve-sbbn.sh predict_sterile_ nu_xray_line	{"m_s_eV": 29000, "sin2_ 2theta": 2.022e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 29 keV, sin\u00b22\u03b8 = 2e-11\n\nE_\u03b3 = 14.5 keV\nDecay ...
71	serve-sbbn.sh predict_sterile_ nu_xray_line	{"m_s_eV": 20000, "sin2_ 2theta": 3.946e-11, "target": "M31"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for M31\nm_s = 20 keV, sin\u00b22\u03b8 = 3.9e-11\n\nE_\u03b3 = 10 keV\nDecay rate = 1.72e-26 s\u207b\u00b9\ nLifetime = 5.82e+25 ...
72	local download_arxiv_ full_text	{"arxiv_id": "astro-ph/ 0511630", "output_dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/runs/ 261019_143114"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/ runs/261019_143114/ astro-ph_0511630.txt
73	local get_arxiv_ metadata	{"arxiv_id": "1702.01764"}	arXiv ID: 1702.01764 Title: New Constraints on the free-streaming of warm dark matter from intermediate and small scale Lyman-α forest data Authors: Vid Ir?i?, Matteo Viel, Martin G. Haehnelt, James...

#	Tool	Arguments	Outcome
74	local get_arxiv_ metadata	{"arxiv_id": "hep-ph/ 0612182"}	arXiv ID: hep-ph/0612182 Title: Lightest sterile neutrino abundance within the nuMSM Authors: Takehiko Asaka, Mikko Laine, Mikhail Shaposhnikov Year: 2006 Abstract: We determine the abundance of the l...
75	local get_arxiv_ metadata	{"arxiv_id": "1911.09073"}	arXiv ID: 1911.09073 Title: Hints, neutrino bounds and WDM constraints from SDSS DR14 Lyman-α and Planck full-survey data Authors: Nathalie Palanque-Delabrouille, Christophe Yèche, Nils Schöneberg, ...
76	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/ runs/261019_143114/ 1609.00667.txt", "start": 0, "length": 6000}	[chars 0-6000 of 51891] (more available, next start=6000) Title: (Almost) Closing the νMSM Sterile Neutrino Dark Matter Window with NuSTAR Authors: Kerstin Perez, Kenny C. Y. Ng, John F. Beacom, Co...
77	local get_arxiv_ metadata	{"arxiv_id": "0812.0010"}	arXiv ID: 0812.0010 Title: Lyman-alpha constraints on warm and on warm-plus-cold dark matter models Authors: Alexey Boyarsky, Julien Lesgourgues, Oleg Ruchayskiy, Matteo Viel Year: 2008 Abstract: We...
78	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-combo-lyman-wdm/ runs/261019_143114/ 1609.00667.txt", "start": 26000, "length": 12000}	[chars 26000-38000 of 51891] (more available, next start=38000)) =(R2 ? +?2- 2R?? cosψ)1/ 2 is the galactocentric radius,ψ is the opening angle from the GC,? is the line of sight distance from th...

#	Tool	Arguments	Outcome
79	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/astro-ph_0511630.txt", "start": 0, "length": 12000}	[chars 0-12000 of 33951] (more available, next start=12000) Title: Production and Evolution of Perturbations of Sterile Neutrino Dark Matter Authors: Kevork Abazajian Year: 2005 arXiv ID: astro-ph/05...
80	local search_arxiv_abstracts	{"max_results": 8, "query": "Lyman-alpha P1D emulator thermal relic warm dark matter mass limit DESI eBOSS"}	[2203.16563v1] Could fresh lava be (warm) dark matter? Abstract: Dark matter models can be classified according to their impact on the properties of galaxies, including cold dark matter (CDM), warm da...
81	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/astro-ph_0511630.txt", "start": 12000, "length": 8000}	[chars 12000-20000 of 33951] (more available, next start=20000) f the same ?avor as the active neutrino. The asymmetric lepton potential for ?avor α is V L = $\sqrt{}$ 2GF ? ?2 (n$\nu\alpha$ - n$^{-}\nu\alpha$) + $\sum$ β ?=α (n$\nu\beta$ ...
82	hep-research-aid get_paper_figures	{"paper_id": "1609.00667", "figures": "6,7,8", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114"}	{"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:1609.00667 (arxiv-html: requested figures [6, 7, 8] not found on https://arxiv.org/html/1609.00667 (available: 3, 4); ar...
83	local search_arxiv_abstracts	{"max_results": 5, "query": "Bozek sterile neutrino thermal relic mass relation 1.294 transfer function"}	[0507333v2] UHE neutrino damping in a thermal gas of relic neutrinos Abstract: We present a calculation of the damping of an ultra-energetic (UHE) cosmic neutrino travelling through the thermal gas ...

#	Tool	Arguments	Outcome
84	local get_arxiv_ metadata	{"arxiv_id": "2007.13751"}	arXiv ID: 2007.13751 Title: General framework for cosmological dark matter bounds using N-body simulations Authors: Keir K. Rogers, Hiranya V. Peiris Year: 2020 Abstract: We present a general framew...
85	board post_finding	{"claim": "Comparing mappings from thermal-relic mass to NRP/DW mass (my own arithmetic, no tool used): Viel+2005 Eq. 5 (4.43 keV m_x^{4/3}, omega= 0.1225) gives 20.0, 21.8, 23.5, 29.1, 40.9 and 45.1 keV for m_x = 3.1, 3....	Recorded finding f-010.
86	board post_finding	{"claim": "Of the abstracts checked, only Boyarsky+2009 (0812.0010) quotes non-resonant (NRP) sterile-neutrino masses directly: m_NRP > 8 keV (99.7%, frequentist) and > 12.1 keV (Bayesian 95%). Inverting Viel+2005 by han...	Recorded finding f-011.
87	board post_dead_end	{"what": "Two search_ arxiv_abstracts queries for post-2023 Lyman-alpha WDM limits: 'one-dimensional flux power spectrum warm dark matter 2024' and 'Lyman-alpha P1D emulator thermal relic warm dark matter mass limit DESI ...	Recorded dead end x-02.
88	board post_finding	{"claim": "Perez+2017 (arXiv:1609.00667, NuSTAR Galactic Center 0-bounce photons) states in the text that, within the nuMSM, its limit gives an upper bound on the sterile-neutrino mass of m_chi below about 16 keV. The nu...	Recorded finding f-012.

#	Tool	Arguments	Outcome
89	board post_finding	{"claim": "Perez+2017 Eq. 5 gives the GC decay-line flux as F ≈ 2.6e-6 cm^-2 s^-1 (m/20 keV)^4 (sin^2 2theta/1e-14)(dOmega/4 deg^ 2)(J/40 GeV cm^-3 kpc sr^ -1), with dOmega_tot ≈ 3.8 deg^2 and J_tot = 29 GeV cm^ -3 kpc sr^-...	Recorded finding f-013.
90	board post_finding	{"claim": "The sterile-nu tool's X-ray verdicts at the DW relic mixing give only a list of excluding limits, never the limit value itself. At m_s = 20 keV, sin^2 2theta = 3.946e-11, the M31 flux is 0.0171 ph/s/cm^2 and t...	Recorded finding f-014.
91	board post_dead_end	{"what": "I tried get_ paper_figures on arXiv: 1609.00667, Figs. 6, 7 and 8, to read approximate NuSTAR GC sin^2 2theta limits.", "why": "Neither the arXiv HTML nor the ar5iv rendering includes Figs. 6-8 (only 3 and 4). Th...	Recorded dead end x-03.
92	board post_question	{"text": "What are the numerical NuSTAR sin^ 2 2theta upper limits (Perez+2017 Fig. 8, Roach+2020 1908.09037, Roach+2023) at m_s = 29, 41 and 46 keV, given that the 40 keV 'bump' in Perez+2017 could weaken the exclusion o...	Recorded question q-07.
93	board post_finding	{"claim": "Abazajian 2005 (astro-ph/0511630, Eq. 7, full numerical DW calculation with the QCD transition, nu_e mixing) : m_s = 3.40 keV (sin^ 2 2theta/1e-8)^(-0.615) (Omega_DM/0.26)^0.5 x {0.527 erfc[-1.15 (T_QCD/170 MeV)...	Recorded finding f-015.

#	Tool	Arguments	Outcome
94	board post_finding	{"claim": "Literature DW relic mixing from the Abazajian 2005 Eq. 7 (my hand calculation, Omega_DM = 0.26, T_QCD = 170 MeV): sin^ 2 2theta = 1.23e-8 (3 keV), 3.02e-9 (7.1 keV), 5.61e-10 (20 keV), 3.06e-10 (29 keV), 1.75e-...	Recorded finding f-016.
95	board post_finding	{"claim": "Revised DW fractions under the Abazajian 2005 yield, f = (sin^2 2theta / sin^ 2 2theta_relic)^1.23 normalized to Omega_DM = 0.26 (my hand calculation) : at 7.1 keV, f = 0.0064 for sin^2 2theta = 5e-11 and 0.0098...	Recorded finding f-017.
96	board post_finding	{"claim": "Corrected Delta N_eff statement: on the relic curve the sterile number density is fixed by Omega_s h^2 / m_s, whatever yield normalization is used. My hand estimate, assuming one Weyl species and n_s/n_ nu = Om...	Recorded finding f-018.
97	board post_question	{"text": "Does the updated DW yield in Abazajian 2017 (1705.01837) or Venumadhav+2016 move the 7.1 keV relic mixing away from the 2005-fit value of 3.0e-9, and should the team quote a range? A cite() call with the verbat...	Recorded question q-08.

A.5 Files produced

- astro-ph_0501562.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testi projects/ramp-combo-lyman-wdm/runs/261019_143114/astro-ph_0501562.txt
- sterile_nu_grid_DW_4bf99b.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/ hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_ nu_grid_DW_4bf99b.csv
- 2309.04533.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/ projects/ramp-combo-lyman-wdm/runs/261019_143114/2309.04533.txt

- `sterile_nu_grid_DW_8d35ed.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_nu_grid_DW_8d35ed.csv`
- `1702.01764.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/1702.01764.txt`
- `sterile_nu_grid_DW_e67bfc.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_nu_grid_DW_e67bfc.csv`
- `sterile_nu_grid_DW_10ad91.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/sterile_nu_grid_DW_10ad91.csv`
- `scan_DW_d94a06.csv` (csv, scan_sterile_nu_parameters): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/scan_DW_d94a06.csv`
- `scan_DW_d94a06.npz` (other, scan_sterile_nu_parameters): `/Users/nramachandra/hep-genesis-testi/projects/ramp-combo-lyman-wdm/runs/261019_143114/scan_DW_d94a06.npz`
- `exclusion_scan_DW_d94a06.pdf` (pdf, plot_sterile_nu_exclusion): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/exclusion_scan_DW_d94a06.pdf`
- `1609.00667.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testing/projects/ramp-combo-lyman-wdm/runs/261019_143114/1609.00667.txt`
- `astro-ph_0511630.txt` (text, download_arxiv_full_text): `/Users/nramachandra/hep-genesis-testi/projects/ramp-combo-lyman-wdm/runs/261019_143114/astro-ph_0511630.txt`

B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

B.1 Directions

Id	Status	Goal	Outcome
d-01	closed	Use the sterile-nu server to compute non-resonant (Dodelson-Widrow) production fractions Omega_s/Omega_dm and related observables across keV masses and mixing angles. Then locate the DW relic-abundance curve (Omega_s h^2 = 0.12) in the (m_s, sin^2 2theta) plane.	Using the sterile-nu server (serve-sbbn), I ran a DW (non-resonant) grid of 15 masses from 1 to 50 keV times 13 mixings from 1e-13 to 1e-7 (CSV sterile_nu_grid_DW_8d35ed.csv), plus a follow-up grid at the five DW-equivalent masses (sterile_nu_grid_DW_10ad91.csv). On this server Omega_s h^2 is exactly linear in sin^2 2theta and grows roughly as m_s^1.8. Setting Omega_s h^2 = 0.12 gives the DW relic
d-02	closed	Verify the recent Lyman-alpha thermal-relic WDM limits and their systematic variations from the papers' text, and verify the thermal-relic to DW sterile-neutrino mass conversion.	Conversion checked against the paper text. Viel+2005 (astro-ph/0501562), Eq. 5, gives m_s(DW) = 4.43 keV (m_x/1 keV)^{4/3} (0.25*0.7^2/omega_x)^{1/3}, with the omega_x normalization at 0.1225 (tc-...-31; cited in references.bib, tc-...-41). The paper says the masses map one-to-one when the transfer functions match, which assumes a rescaled Fermi-Dirac spectrum. The survey's 4.46 keV prefactor does
d-03	closed	Build the X-ray decay and mixed cold-plus-warm side of the comparison. This means X-ray mixing-angle limits at the DW-equivalent masses, the predicted line energies and fluxes, and the published Lyman-alpha bounds on the warm fraction f versus mass.	Partly done: the X-ray limits above 26 keV and the DW-yield check (step 4) are still open. (1) The sterile-nu server does compute keV-scale DW relic densities (answers q-01). Its DW relic curve (Omega_s h^2 = 0.12) gives sin^2 2theta = 2.545e-10, 3.946e-11, 2.022e-11, 1.084e-11 and 8.81e-12 at m_s = 7.1, 20, 29, 41 and 46 keV, with radiative lifetimes of 1.6e27, 5.82e25, 1.77e25, 5.85e24 and 4.05e

Id	Status	Goal	Outcome
d-04	blocked	Extract numerical NuSTAR X-ray upper limits on sin^2 2theta at m_s = 20, 29, 41 and 46 keV. Combine them with the DW relic curve to get the X-ray-allowed fraction f_X(m_s), compare f_X with the Boyarsky+2009 mixed cold-plus-warm bound, and give a final verdict on pure non-resonant production over 20-46 keV.	I could not get numerical NuSTAR sin^2 2theta limits at 20, 29, 41 or 46 keV, so there is no f_X(m_s) table and no f_X comparison with the Boyarsky+2009 bound (F_WDM < 0.35-0.6). Pure DW comes out excluded across 20-46 keV, but only by inference from the Perez+2017 text, not from numbers. What was established: (1) Perez+2017 (arXiv:1609.00667; full text in run_dir/1609.00667.txt, tc-...-69, -76,
d-05	closed	Cross-check the sterile-nu server's DW yield normalization and its sin^2 2theta scaling against published production calculations. Derive a correction factor and propagate it to the relic curve (f-003), the fractions f (f-004) and the Delta N_eff statement.	Two success criteria are met (relic-mixing ratios at six masses, revised fractions f with the normalization stated) and the corrected Delta N_eff statement is recorded as an analytic estimate. The cite() step was not done. All numbers below except the fit itself are my hand arithmetic from that fit, not tool output. (1) Literature yield (f-015). Abazajian 2005 (astro-ph/0511630, Eq. 7), read verb
d-06	closed	Test how robust the thermal-relic to NRP mass mapping is by tabulating paper-quoted NRP-equivalent masses next to the Viel+2005-derived values. Run one targeted search for post-2023 Lyman-alpha WDM limits from DESI P1D or emulator re-analyses.	I built a side-by-side table of NRP/DW-equivalent masses under two relations (finding f-010). I did the arithmetic by hand; no tool computed it. The relations are Viel+2005 Eq. 5 (4.43 keV m_x^{4/3}, omega = 0.1225; verified in d-02) and a Bozek+2016-type relation (3.9 keV m_x^{1.294}). I took the Bozek coefficients from the direction card and could not confirm them in any paper text: the abstract

B.2 Findings and evidence

Id	Confidence	Claim	Evidence (tool calls, figures)
f-001	medium	On the sterile-nu server's DW relic curve (Omega_s h^2 = 0.12), the mixing is sin^2 2theta = 2.545e-10, 3.946e-11, 2.022e-11, 1.084e-11 and 8.81e-12 at m_s = 7.1, 20, 29, 41 and 46 keV. The corresponding radiative lifetimes are 1.6e27, 5.82e25, 1.77e25, 5.85e24 and 4.05e24 s.	tc-261019_1-w1-d-03-26, tc-261019_1-w1-d-03-35
f-002	medium, **contested**	At the DW relic mixing, the line energies are E = 3.55, 10, 14.5, 20.5 and 23 keV for m_s = 7.1, 20, 29, 41 and 46 keV. The whole-halo M31 fluxes are 1.75e-3, 1.71e-2, 3.88e-2, 8.31e-2 and 0.107 ph/s/cm^2. The tool's tabulated limits exclude m_s = 7.1 keV (NuSTAR M31 Roach+2020 and XMM blank sky Dessert+2020) and m_s = 20 keV (NuSTAR M31). They report no tabulated limit within 30% in mass at 29, 41 or 46 keV. That means no coverage, not that these masses are allowed.	tc-261019_1-w1-d-03-36, tc-261019_1-w1-d-03-37, tc-261019_1-w1-d-03-38, tc-261019_1-w1-d-03-39, tc-261019_1-w1-d-03-40
f-003	high	On the sterile-nu server (DW mechanism), Omega_s h^2 scales linearly with sin^2 2theta and roughly as m_s^1.8. Setting Omega_s h^2 = 0.12 gives a DW relic curve of sin^2 2theta ≈ 8.7e-9 (1 keV), 2.5e-9 (2 keV), 1.2e-9 (3 keV), 4.8e-10 (5 keV), 2.5e-10 (7.1 keV), 1.4e-10 (10 keV), 6.6e-11 (15 keV), 3.9e-11 (20 keV), 3.3e-11 (22 keV), 2.0e-11 (29 keV), 1.08e-11 (41 keV), 8.8e-12 (46 keV) and 7.6e-12 (50 keV).	tc-261019_1-w1-d-01-29, tc-261019_1-w1-d-01-34, tc-261019_1-w1-d-01-47
f-004	high, **contested**	DW fractions f = Omega_s h^2 / 0.12 from the server: at 7.1 keV and sin^2 2theta = 5e-11, f = 0.196 (Omega_s h^2 = 0.0236). At sin^2 2theta = 1e-12, f = 0.025 (20 keV), 0.030 (22 keV), 0.049 (29 keV), 0.092 (41 keV) and 0.114 (46 keV). At 1e-11, f is ten times these values; 46 and 50 keV then overclose (Omega_s h^2 = 0.136 and 0.158). The free-streaming length is 151 kpc at 7.1 keV, 53.7 kpc at 20 keV and 107 kpc at 10 keV. Delta N_eff is at most 3e-4 and BBN is unconstrained for any DW keV dark matter near the relic curve.	tc-261019_1-w1-d-01-24, tc-261019_1-w1-d-01-47, tc-261019_1-w1-d-01-34

Id	Confidence	Claim	Evidence (tool calls, figures)
f-005	high	Verified from the text of Viel+2005 (astro-ph/0501562, Eq. 5): m_s(DW) = 4.43 keV (m_x/1 keV)^{4/3} (0.25*0.7^2/omega_x)^{1/3}, normalized at omega_x = 0.1225; the survey's 4.46 keV prefactor is not in the paper but equals 4.43*(0.1225/0.12)^{1/3} = 4.461 keV once rescaled to omega_dm = 0.12. DW-equivalent masses (omega_dm=0.1225 / 0.12): m_x=3.1 -> 20.0/20.2 keV; 3.3 -> 21.8/21.9; 3.5 -> 23.5/23.7; 4.1 -> 29.1/29.3; 4.5 (Villasenor peak) -> 32.9/33.1; 5.3 -> 40.9/41.2; 5.7 -> 45.1/45.4 keV (m_x^{4/3} = 4.520, 4.913, 5.314, 6.562, 7.430, 9.241, 10.182). The survey's range of 20-46 keV is confi	tc-261019_1-w1-d-02-31, tc-261019_1-w1-d-02-41
f-006	high	Verified Lyman-alpha thermal-relic WDM limits: Viel+2013 (HIRES/MIKE, z>4) m_x>3.3 keV (2 sigma; 1, 2, 2.5 keV disfavoured at about 9, 4, 3 sigma); Irsic+2017 XQ-100 alone 1.4 keV, HIRES/MIKE 4.1 keV, combined 5.3 keV (2 sigma), falling to 3.5 keV if sudden temperature jumps of up to 5000 K are allowed; Palanque-Delabrouille+2019 SDSS DR14 z<4.5 plus XQ-100 m_X>5.3 keV (95% CL); Villasenor+2022 (1080 Cholla sims, Boera+2019 data, z=4.0-5.2) m_WDM>3.1 keV (95% CL) with a likelihood peak at 4.5 keV and FPS suppression saturating at k>~0.1 s/km from peculiar-velocity smearing; Irsic+2023 HIRES/UV	tc-261019_1-w1-d-02-32, tc-261019_1-w1-d-02-33, tc-261019_1-w1-d-02-42, tc-261019_1-w1-d-02-43, tc-261019_1-w1-d-02-44, ...
f-007	high	The mixed cold-plus-warm Lyman-alpha bound of Boyarsky+2009 (arXiv:0812.0010), with masses expressed as non-resonantly produced sterile-neutrino masses, allows any mass in 5 keV < m_NRP < infinity when F_WDM < 0.6 (frequentist 99.7% CL) or F_WDM < 0.35 (Bayesian 95%). The pure-WDM bounds are m_NRP > 8 keV (99.7%) and 12.1 keV (Bayesian 95%).	tc-261019_1-w1-d-03-54, tc-261019_1-w1-d-03-55
f-008	medium	NuSTAR Galactic Center observations (Perez+2017, arXiv:1609.00667) give, by the abstract's own description, the strongest X-ray line limits on sterile-neutrino mixing over most of 10-50 keV in mass, improving earlier limits by up to about 10x. This is the published limit set that covers the DW-equivalent masses of 29-46 keV, which the server's tabulated limits do not cover.	tc-261019_1-w1-d-03-53, tc-261019_1-w1-d-03-51

Id	Confidence	Claim	Evidence (tool calls, figures)
f-009	high	I made a DW exclusion plot over m_s = 1-100 keV and sin^2 2theta = 1e-14 to 1e-6, from an 80x80 scan (6400 points): 3092 points overclose and 968 are BBN-excluded. The plot shows the overclosure region, the Omega_DM line, the NuSTAR/XMM limits, the Lyman-alpha thermal-WDM bound and the 3.5 keV point. It is Figure 10.	tc-261019_1-w1-d-01-48, tc-261019_1-w1-d-01-58 Fig. 10
f-010	medium, **contested**	Comparing mappings from thermal-relic mass to NRP/DW mass (my own arithmetic, no tool used): Viel+2005 Eq. 5 (4.43 keV m_x^{4/3}, omega=0.1225) gives 20.0, 21.8, 23.5, 29.1, 40.9 and 45.1 keV for m_x = 3.1, 3.3, 3.5, 4.1, 5.3 and 5.7 keV. A Bozek+2016-type relation, 3.9 keV m_x^{1.294}, gives 16.9, 18.3, 19.7, 24.2, 33.8 and 37.1 keV. That is lower by 3.1, 3.5, 3.8, 4.9, 7.1 and 8.0 keV, so the DW-equivalent range moves from 20-45 keV to 17-37 keV. The 3.9 keV and 1.294 coefficients come from the direction card. I could not confirm them from any paper text: the abstract search found nothing, a	tc-261019_1-w2-d-06-82, tc-261019_1-w2-d-06-84, tc-261019_1-w1-d-02-31
f-011	high	Of the abstracts checked, only Boyarsky+2009 (0812.0010) quotes non-resonant (NRP) sterile-neutrino masses directly: m_NRP > 8 keV (99.7%, frequentist) and > 12.1 keV (Bayesian 95%). Inverting Viel+2005 by hand, these correspond to thermal-relic masses of about 1.56 and 2.12 keV, which are older and weaker than the 3.1-5.7 keV limits. The abstracts of Irsic+2017 (1702.01764) and Palanque-Delabrouille+2019 (1911.09073) give only thermal-relic masses (5.3 keV for both; 3.5 keV for Irsic+2017 with temperature jumps), so the DW-equivalent values for them come entirely from a mapping relation.	tc-261019_1-w2-d-06-77, tc-261019_1-w2-d-06-71, tc-261019_1-w2-d-06-75

Id	Confidence	Claim	Evidence (tool calls, figures)
f-012	medium, **contested**	Perez+2017 (arXiv:1609.00667, NuSTAR Galactic Center 0-bounce photons) states in the text that, within the nuMSM, its limit gives an upper bound on the sterile-neutrino mass of m_chi below about 16 keV. The nuMSM allowed band in its Fig. 8 lies between the non-resonant (DW) production line above and the maximum-lepton-asymmetry resonant line below. So the NuSTAR GC limit lies below the whole band, including the DW line, at masses above about 16 keV and up to the edge of its 10-50 keV coverage. Above 50 keV, INTEGRAL limits rule out mixing-produced sterile neutrinos as all of the dark matter. T	tc-261019_1-w2-d-04-78, tc-261019_1-w2-d-04-76
f-013	medium, **contested**	Perez+2017 Eq. 5 gives the GC decay-line flux as F ≈ 2.6e-6 cm^-2 s^-1 (m/20 keV)^4 (sin^2 2theta/1e-14)(dOmega/4 deg^2)(J/40 GeV cm^-3 kpc sr^-1), with dOmega_tot ≈ 3.8 deg^2 and J_tot = 29 GeV cm^-3 kpc sr^-1 for the default coreNFW profile. Applied to the DW relic mixing at 20 keV (3.9e-11), this predicts about 7e-3 ph/cm^2/s in the NuSTAR 0-bounce aperture (my hand calculation). Numerical flux and sin^2 2theta limits are shown only in Figs. 6-8, which are not machine-readable here.	tc-261019_1-w2-d-04-78
f-014	high	The sterile-nu tool's X-ray verdicts at the DW relic mixing give only a list of excluding limits, never the limit value itself. At m_s = 20 keV, sin^2 2theta = 3.946e-11, the M31 flux is 0.0171 ph/s/cm^2 and the result is 'EXCLUDED by: NuSTAR M31 (Roach+ 2020)'. At m_s = 29 keV, sin^2 2theta = 2.022e-11 (E = 14.5 keV, tau = 1.77e25 s, M31 flux 0.0388, MW-center whole-halo flux 264 ph/s/cm^2), the result is 'Not excluded by tabulated X-ray limits (within 30% in mass)' with excluded_by = []. So f-002 cannot be backed by a numerical limit from this tool, and its 29 keV 'not excluded' is a coverag	tc-261019_1-w2-d-04-72, tc-261019_1-w2-d-04-73

Id	Confidence	Claim	Evidence (tool calls, figures)
f-015	high, **contested**	Abazajian 2005 (astro-ph/0511630, Eq. 7, full numerical DW calculation with the QCD transition, nu_e mixing): m_s = 3.40 keV (sin^2 2theta/1e-8)^(-0.615) (Omega_DM/0.26)^0.5 x {0.527 erfc[-1.15 (T_QCD/170 MeV)^2.15]}. The braced factor is 1 at T_QCD = 170 MeV, and the fit is valid for 135 < T_QCD < 300 MeV. Inverted, this gives Omega_s = 0.26 (m_s/3.4 keV)^2 (sin^2 2theta/1e-8)^1.23. Compared with the server's linear scaling in mixing and roughly m^1.8 in mass, the literature yield goes as mixing^1.23 and m_s^2. The peak production temperature is T_peak ~ 130 MeV (m_s/3 keV)^(1/3).	tc-261019_1-w2-d-05-81, tc-261019_1-w2-d-05-70
f-016	medium, **contested**	Literature DW relic mixing from the Abazajian 2005 Eq. 7 (my hand calculation, Omega_DM = 0.26, T_QCD = 170 MeV): sin^2 2theta = 1.23e-8 (3 keV), 3.02e-9 (7.1 keV), 5.61e-10 (20 keV), 3.06e-10 (29 keV), 1.75e-10 (41 keV), 1.45e-10 (46 keV). The server's relic curve (f-003) lies 10.2x, 12.1x, 14.4x, 15.3x, 16.1x and 16.4x lower in mixing at these masses. So the server overproduces DW sterile neutrinos at fixed mixing by more than the critic's 2.5x estimate. The f-001 and f-003 relic mixings should be multiplied by about 10-16 to match this fit. Using Omega_DM = 0.264 (h = 0.674) raises the lite	tc-261019_1-w2-d-05-81
f-017	medium, **contested**	Revised DW fractions under the Abazajian 2005 yield, f = (sin^2 2theta / sin^2 2theta_relic)^1.23 normalized to Omega_DM = 0.26 (my hand calculation): at 7.1 keV, f = 0.0064 for sin^2 2theta = 5e-11 and 0.0098 for 7e-11. This matches the literature statement of about 1% of the dark matter at the 3.5 keV-line mixing, whereas the server gives 0.196 and 0.27 (about 30x higher). At sin^2 2theta = 1e-12: f = 4.2e-4 (20 keV), 8.8e-4 (29 keV), 1.75e-3 (41 keV), 2.2e-3 (46 keV), against the server's 0.025, 0.049, 0.092 and 0.114 (50-60x too high). At 1e-11, multiply by 10^1.23 = 17: f = 7e-3 (20 keV)	tc-261019_1-w2-d-05-81

Id	Confidence	Claim	Evidence (tool calls, figures)
f-018	medium	Corrected Delta N_eff statement: on the relic curve the sterile number density is fixed by Omega_s h^2 / m_s, whatever yield normalization is used. My hand estimate, assuming one Weyl species and n_s/n_nu = Omega_s h^2 x 94 eV / m_s, gives Delta N_eff ≈ 11.3 eV / m_s while relativistic: 1.6e-3 at 7.1 keV, 5.6e-4 at 20 keV, 2.5e-4 at 46 keV. This matches the critic's 1.6e-3 on-curve value at 7.1 keV, so f-004's 'at most 3e-4' holds only above about 40 keV. All values are 2-3 orders of magnitude below BBN and Planck sensitivity (about 0.2-0.3), so BBN does not constrain keV DW dark matter.	tc-261019_1-w2-d-05-81

B.3 Dead ends

- x-01 (d-02): A search_arxiv_abstracts query for a post-2023 DESI Lyman-alpha thermal-relic WDM limit — The query returned only DESI 2024 BAO and sample papers. No DESI DR1 P1D-based WDM mass limit came up, so no newer limit was added. A query on P1D or 'one-dimensional flux power' might still find one.

- x-02 (d-06): Two search_arxiv_abstracts queries for post-2023 Lyman-alpha WDM limits: 'one-dimensional flux power spectrum warm dark matter 2024' and 'Lyman-alpha P1D emulator thermal relic warm dark matter mass limit DESI eBOSS'. — Neither query returned a P1D-based WDM limit from DESI DR1, eBOSS or an emulator re-analysis. The results were unrelated papers plus DESI BAO papers (2404.03001, 2503.14739). This search tool appears to rank by keywords poorly. Next step: web_search, or arxiv_agent with author names (Rogers, Irsic, Karacayli, Ravoux, Cabayol-Garcia).

- x-03 (d-04): I tried get_paper_figures on arXiv:1609.00667, Figs. 6, 7 and 8, to read approximate NuSTAR GC sin^2 2theta limits. — Neither the arXiv HTML nor the ar5iv rendering includes Figs. 6-8 (only 3 and 4). The limits appear only in the figures, and the text gives no tabulated values. Reading them needs the PDF figure, or the Roach+2020 (1908.09037) or Roach+2023 data releases.

B.4 Critic verdicts

- c-01 (wave 0, plan): **revise**; high: The sterile-nu server is serve-sbbn.sh. Its probes (the eV-scale SBL anomaly check, BBN Yp and D/H, N_eff) suggest it may only model thermalized light steriles, not keV Dodelson-Widrow freeze-in. Make step 1 a hard gate. If describe_sterile_nu_tools or a single predict_sterile_nu call at 7.1 keV doe; medium: Add a robustness cross-check. Compare the tool's DW relic curve at 2-3 masses against a published analytic DW abundance, for example the Abazajian/Asaka fits in 1705.01837. Without this, f values tied only to tool calls cannot be verified. Also note that predict_sterile_nu_grid and scan_sterile_nu_p; high: This does not fit 13 calls. Six papers, each needing download, chunk reads and cite, plus a post-2023 search, needs about 20 or more calls. Prioritize Viel+2005 for the conversion and 2309.04533 and 1702.01764 for the limits and systematics, all with full text. Take 2209.14220, 1911.09073 and

1306.2; medium: Several of these papers quote their own NRP sterile-neutrino-equivalent masses. Irsic+2017 and Palanque-Delabrouille+2019 are examples, and they use updated conversions that differ from the Viel+2005 4/3 power law. Report both the paper-quoted value and the formula-derived value side by side, rather; medium: Step 1 runs X-ray line predictions with survey-estimated mixing angles before d-01 delivers the relic curve. Those results would rest on unverified inputs. Either make the step depend on d-01's output, or use one predict_sterile_nu_grid call that returns the X-ray fields rather than five separate ca; low: The scope is broad for 13 calls: X-ray limits, mixed-fraction bounds, and DW yield revisions. Limit full-text reads to one X-ray limit compilation (1702.08430 or a recent NuSTAR paper) and one mixed-model Lyman-alpha paper (Baur+2017 or Boyarsky+2009). Handle 1512.05369 and 2006.09553 from abstracts

- `c-02` (wave 1, wave): **continue**; contested `f-002`, `f-004`; high: The cross-check against a published DW yield requested in c-01 was not done, and the server normalization looks off. The Abazajian (2005) fit, Omega_s h^2 ~ 0.1 (sin^2 2theta/3e-9)^1.23 (m_s/3 keV)^1.8, needs about 3e-9 at 3 keV for the full relic abundance. The server gives 1.2e-9, a factor of abou; low: The lifetime values in f-001 are not in its cited evidence (tc-26, tc-35); they appear only in tc-36 to tc-40. In addition, 3 of the 5 points at Omega_s h^2 = 0.12 are flagged 'overcloses', which shows the overclosure threshold is applied with a strict or rounded inequality. Points sitting exactly o; medium: The verdict on pure DW at 29-46 keV (the Lyman-alpha-allowed masses under the stronger limits) still rests on NuSTAR numbers that have not been extracted (q-06). Without them, the headline 'pure DW excluded' is shown only for the conservative 20 keV end. tc-51 (a two-sterile-neutrino oscillation pap; medium: The arithmetic is correct: 4.43 x 3.1^{4/3} = 20.0 keV and 4.43 x 5.7^{4/3} = 45.1 keV. However, the visible excerpt of tc-31 shows only the title page, not Eq. 5. More importantly, only the Viel+2005 relation was used. Later relations, such as the Bozek+2016-type m_s ~ 3.9 keV (m_x/keV)^1.294 and t; low: The provenance of the NuSTAR/XMM and Lyman-alpha curves overlaid in Figure 10 is not documented. The tool's built-in limits have known gaps at 29-46 keV, so the figure should state which limit set is drawn and where coverage ends.; medium: q-04 is unresolved: the X-ray-allowed f at each mass has not been combined with the Boyarsky+2009 mixed bound. Two further points are needed for a correct comparison. First, Boyarsky's m_NRP uses its own NRP mapping. Second, a DW component with f << 0.35 is trivially allowed by Lyman-alpha, so the m

- `c-03` (wave 2, wave): **continue**; contested `f-010`, `f-012`, `f-013`, `f-015`, `f-016`, `f-017`; high: Recompute the fractions with f linear in mixing at fixed mass, using the literature relic mixing. The 1.23 exponent biases f low by a factor of 2.6-4.; high: There is still no numerical X-ray limit at 29-46 keV. The corrected literature DW relic mixing is 10-16x larger, which makes X-ray exclusion of pure DW much easier: the predicted 20 keV GC flux rises from about 7e-3 to about 0.1 ph/cm^2/s. The verdict should be restated against the corrected relic c; medium: The tool labels its limit 'NuSTAR M31 (Roach+ 2020)'. Roach+2020 (1908.09037) is a Galactic-bulge analysis, while the NuSTAR M31 limit is Ng+2019 (1901.01262), so the provenance needs checking. The 29 keV excerpt (tc-72) is also truncated before the 'not excluded' verdict string.; medium: f-001 and f-003 remain marked supported as server outputs. The board should state explicitly that they use the server normalization and sit 10-16x below literature DW mixings before any final comparison relies on them.; low: The tc-82 evidence is an irrelevant search result and should not be cited as evidence.; low: The Delta N_eff estimate assumes an active-neutrino momentum shape. A <p/T> factor of about 0.9 changes it by about 10%, which does not affect the BBN conclusion.

- `c-04` (wave 2, report): **revise**; high: unsupported: Figure extraction for Perez+2017

Figs. 6–8 failed. Neither the arXiv HTML nor the ar5iv rendering contains them, only Figs. 3 — The report is truncated here, mid-sentence in the dead-ends list. As a result it has no explicit open-directions section. Several board open items are never pr; high: unsupported: Two searches for post-2023 limits from DESI DR1 P1D, eBOSS updates or emulator re-analyses found nothing. — This is internally inconsistent with the dead-ends text, which says three searches. The board records three queries: one in x-01 and two in x-02.; high: unsupported: The critic accepted the arithmetic, but treats the Bozek-type column as contested because its source is unknown. — c-02 accepted the arithmetic for the Viel+2005 relation only. c-03 lists f-010 as contested but gives no stated reason. Attributing the critic's position to the unknown sou; high: unsupported: A 15-mass × 13-mixing DW grid was run on the server, covering 1–50 keV and $\sin^2 2\theta$ = 1e-13 to 1e-7 (files 'sterile_nu_grid_DW_8d35ed.csv' and 'sterile_nu_grid_DW_10ad91.csv'). — Per d-01, only 8d35ed is the 15×13 grid. 10ad91 is a separate follow-up grid at the five DW-equivalent masses.